**19** Two fixed point charges +2Q and -Q are separated by a distance $2d$.

A particle of mass $m$ and charge $+q$ approaches from a large distance with speed $u$. It subsequently passes through the midpoint M between the two fixed charges.

Assume that the fixed charges remain stationary, what is the speed of the particle as it passes through M?

**A** $\sqrt{\frac{Qq}{2\pi\varepsilon_0 md} - u^2}$

**B** $\sqrt{u^2 - \frac{Qq}{2\pi\varepsilon_0 md}}$

**C** $\sqrt{\frac{Qq}{4\pi\varepsilon_0 md} - u^2}$

**D** $\sqrt{u^2 - \frac{Qq}{4\pi\varepsilon_0 md}}$